

BigQuery 對話式數據分析簡介

來源：<https://codelabs.developers.google.com/ca-in-bigquery?hl=zh-tw>

步驟數：8

在本程式碼研究室中，您將瞭解如何使用 BigQuery 的代理程式目錄，建構、調整及發布對話式數據分析資料代理程式。瞭解如何使用 Gemini 生成中繼資料、提供經過驗證的查詢來提升代理程式準確度，以及在 BigQuery 中與業務使用者共用代理程式。

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1. 簡介

從資料取得洞察資料通常需要大量時間、心力，以及深厚的 SQL 專業知識。在本程式碼實驗室中，您將探索 **BigQuery** 的代理程式目錄，這個新平台可透過對話式資料代理程式，即時提供 AI 輔助的洞察資訊。

您將建立精選資料代理程式，進一步瞭解如何將文字轉換為 SQL。您將瞭解如何提供業務脈絡、系統指令和已驗證的查詢，讓代理程式產生高度準確的結果。最後，您將發布這個代理，供貴機構的其他使用者使用。

必要條件

- 對 Google Cloud 有基本瞭解

課程內容

- 如何瀏覽 BigQuery 代理程式目錄
- 如何建立自訂代理程式及定義知識來源
- 如何使用 Gemini 生成語意中繼資料
- 如何新增系統指令和已驗證的查詢，引導代理
- 如何發布及分享代理程式

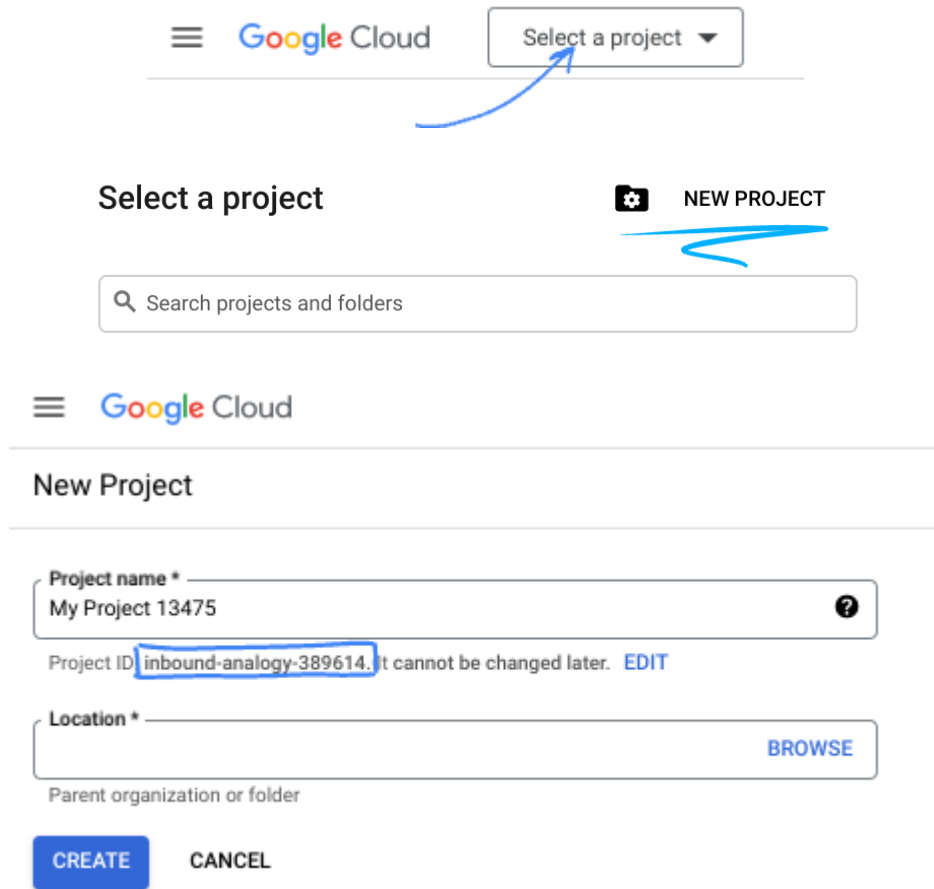
軟硬體需求

- Google Cloud 帳戶和 Google Cloud 專案
 - BigQuery 和 SQL 的基本知識
 - 網路瀏覽器，例如 [Chrome](#)
-

2. 設定和需求

選擇專案

1. 登入 [Google Cloud 控制台](#)，然後建立新專案或重複使用現有專案。如果沒有 Gmail 或 Google Workspace 帳戶，請先[建立帳戶](#)。



The screenshot shows the Google Cloud console interface for selecting or creating a project. At the top, there's a 'Select a project' dropdown menu. Below it, a search bar labeled 'Search projects and folders' is present. To the right, there's a 'NEW PROJECT' button. The 'New Project' form is shown below, with fields for 'Project name *', 'Project ID' (which is 'inbound-analogy-389614' and highlighted with a blue box), 'Location *', and 'Parent organization or folder'. The 'CREATE' button is highlighted with a blue box.

- **專案名稱**是這個專案參與者的顯示名稱。這是 Google API 未使用的字元字串。你隨時可以更新。
- **專案 ID** 在所有 Google Cloud 專案中都是不重複的，而且設定後即無法變更。Cloud 控制台會自動產生專屬字串，通常您不需要在意該字串為何。在大多數程式碼研究室中，您需要參照專案 ID (通常標示為 `PROJECT_ID`)。如果您不喜歡產生的 ID，可以產生另一個隨機 ID。你也可以嘗試使用自己的名稱，看看是否可用。完成這個步驟後就無法變更，且專案期間會維持不變。
- 請注意，有些 API 會使用第三個值，也就是「專案編號」。如要進一步瞭解這三種值，請參閱[說明文件](#)。

注意：專案 ID 在全域中是唯一的，選定後就無法再供他人使用。您是該 ID 的唯一使用者。即使專案已刪除，ID 也無法再次使用

注意：如果使用 Gmail 帳戶，可以將預設位置設為「沒有機構」。如果你使用 Google Workspace 帳戶，請選擇適合貴機構的位置。

2. 接著，您需要在 Cloud 控制台中[啟用帳單](#)，才能使用 Cloud 資源/API。完成本程式碼研究室的費用不高，甚至可能完全免費。如要關閉資源，避免在本教學課程結束後繼續產生費用，請刪除您建立的資源或專案。Google Cloud 新使用

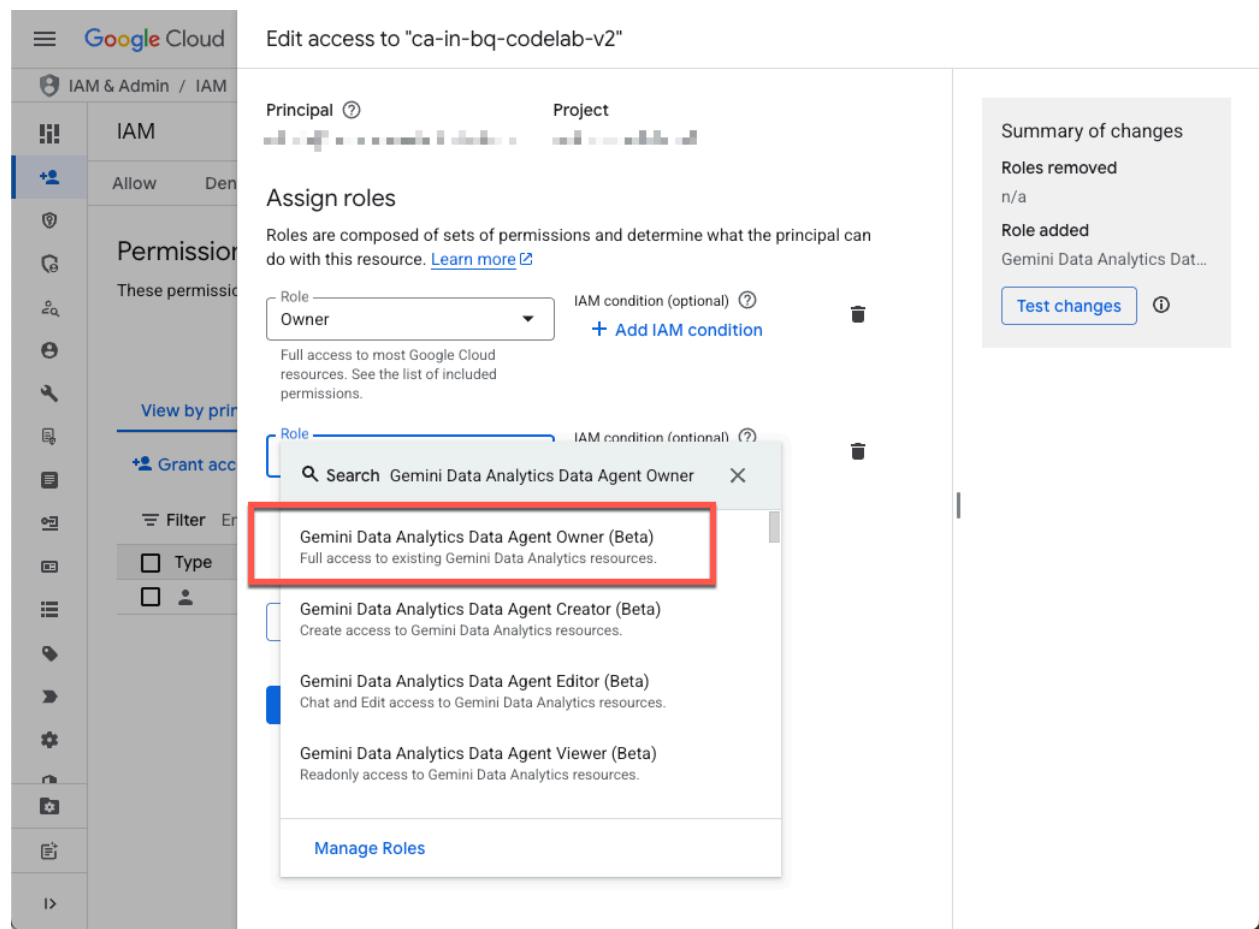
者可參加價值\$300 美元的免費試用計畫。

步驟 3 · 約 5 分鐘

3. 事前準備

授予自己必要角色

前往[專案的 IAM 頁面](#)，然後授予自己 **Gemini Data Analytics** 資料代理擁有者角色：

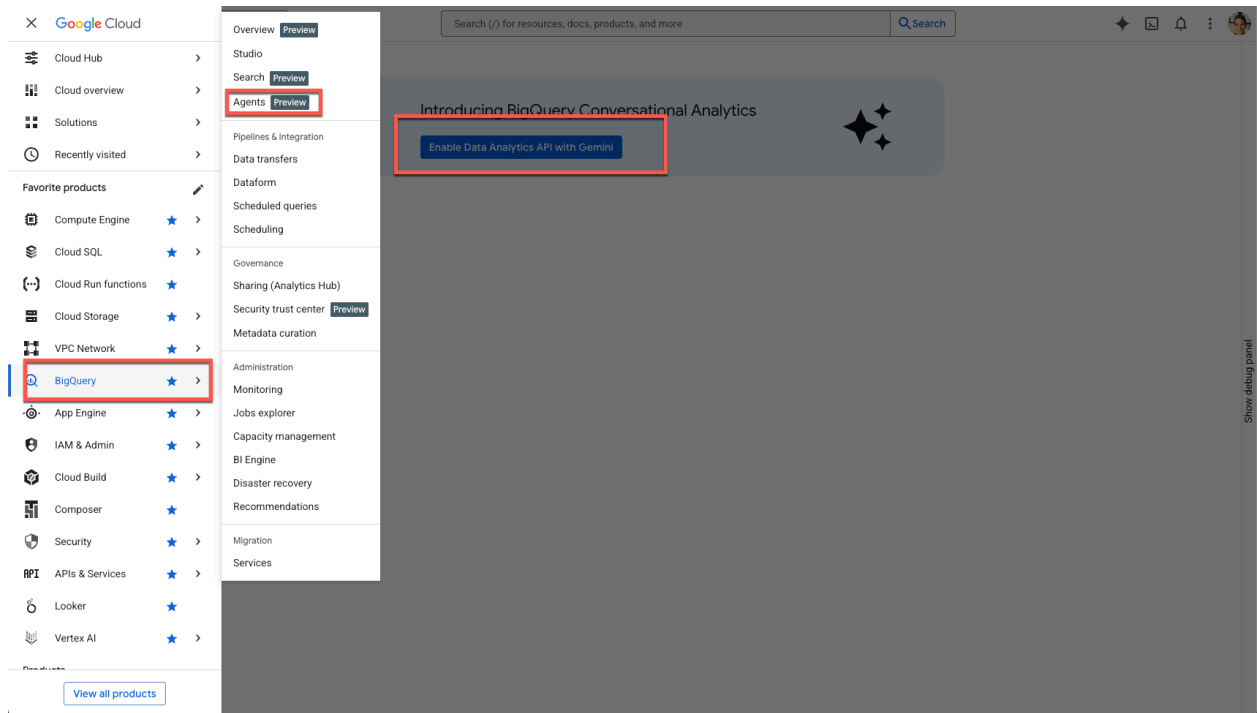


這個角色授予的權限可讓您建立、編輯、共用及刪除專案中的所有資料代理程式。

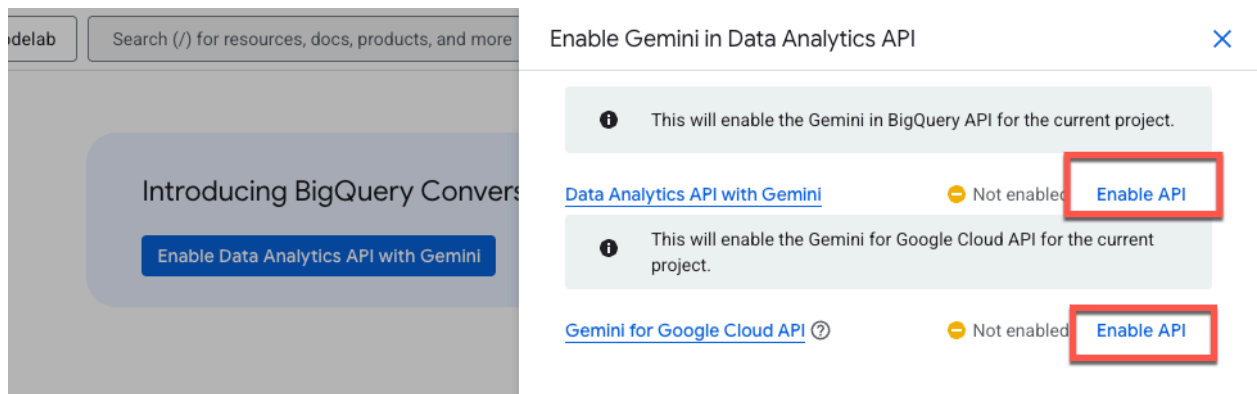
啟用必要的 API

使用側邊導覽選單或頁面頂端的搜尋選單，前往「BigQuery」>「Agents」。

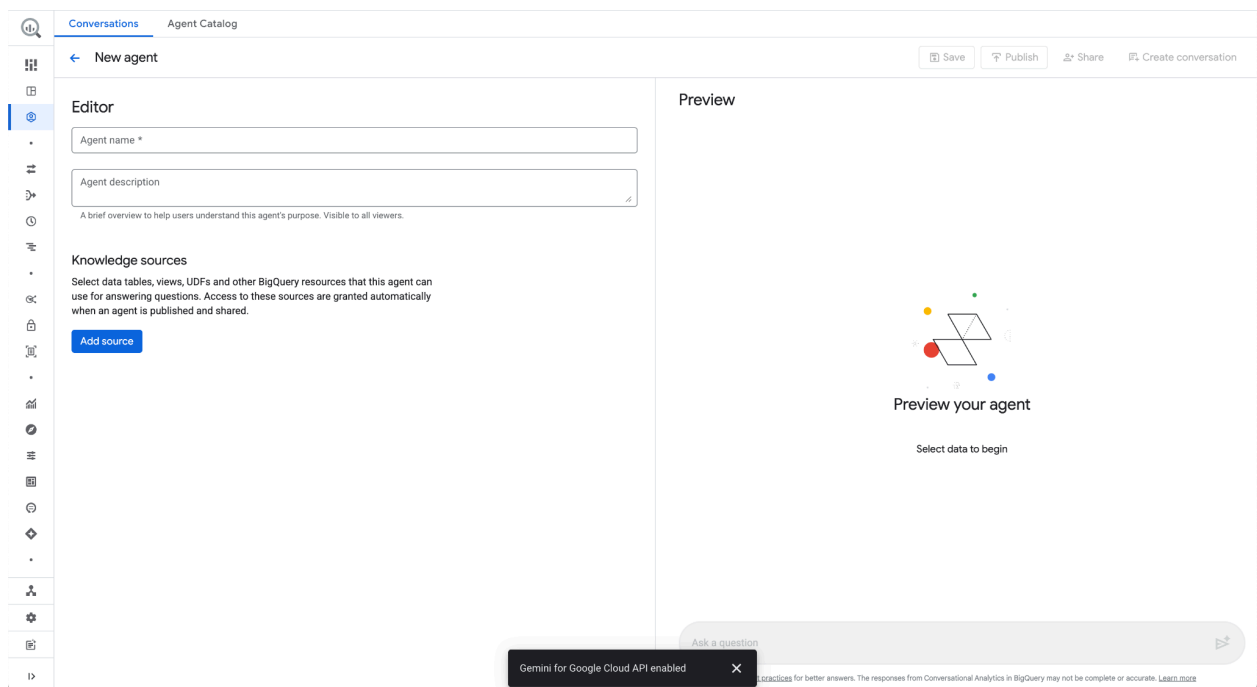
按一下「啟用 Data Analytics API with Gemini」：



同時啟用 **Gemini in BigQuery API** 和 **Gemini for Google Cloud API**：



現在應該會看到新的代理程式頁面：



4. 建立虛擬服務專員

我們將使用 [Google 搜尋趨勢國際公開資料集](#)，建立第一個資料代理程式。這個資料集有助於瞭解全球熱門搜尋字詞，以及這些興趣與過去的比較。

首先，請為代理命名並提供簡要說明。這段說明僅供其他使用者瞭解代理的用途。

代理程式名稱

Google Trends Agent

代理說明

Data agent for the Google Trends International Top Terms public dataset

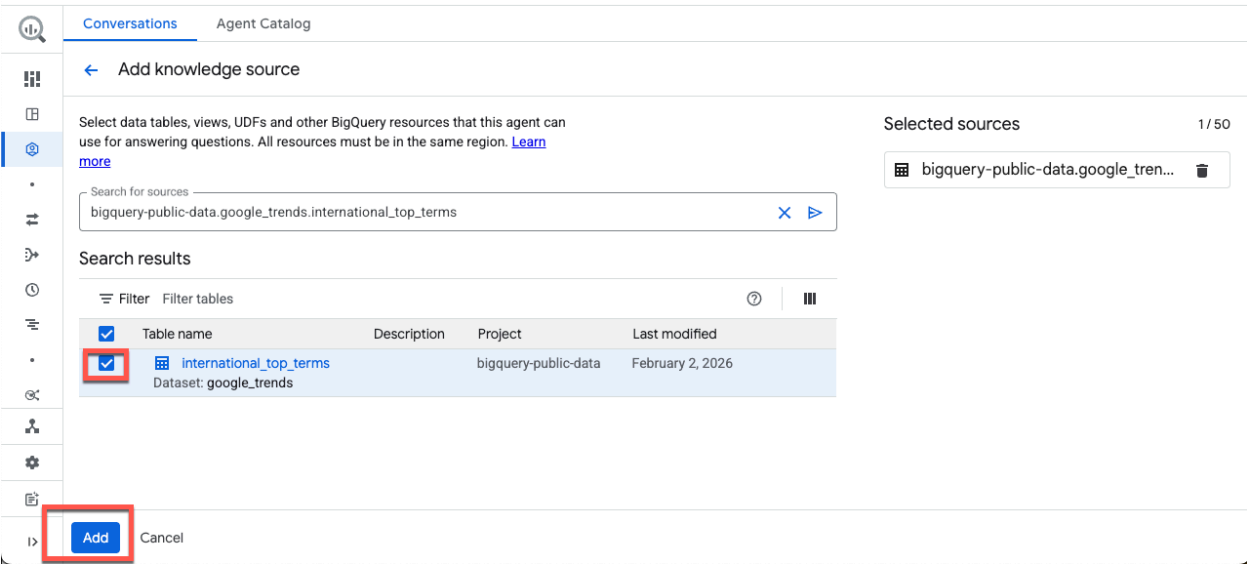
知識來源

接著新增知識來源。知識來源是代理可用來回答問題的 BigQuery 資料表、檢視區塊或 UDF。

為簡化流程，請在本範例中只新增一個表格。不過請注意，每個代理程式最多可新增 50 個知識來源，以處理更複雜的資料情境。

在搜尋框中輸入下表，勾選方塊，然後按一下「新增」：

bigquery-public-data.google_trends.international_top_terms



結構化脈絡

如要提高資料代理的準確率，請在資料表和資料欄中加入結構化脈絡。按一下「自訂」 **Customise**：

Knowledge sources

Select data tables, views, UDFs and other BigQuery resources that this agent can use for answering questions. Access to these sources are granted automatically when an agent is published and shared.

 bigquery-public-data.google_trends.international_top_terms [Customise](#) 

[Manage sources](#)

注意：這項功能可讓您在資料表和資料欄層級新增說明。由於來源是無法直接修改的公開資料集，這些覆寫有助於代理程式瞭解資料，但不會變更原始表格。

Gemini 會自動生成說明建議。按一下「資料表說明」旁的「接受」：

[← Customise](#) [↻ Regenerate suggestions](#)

 Changes made here will not affect the source table. Edit the source table to make global changes. [Edit source table](#)

 bigquery-public-data.google_trends.international_top_terms

Edit table and field descriptions, add synonyms and more to improve your agent's accuracy.

Description

Table description

 Suggestion generated by Gemini

The international_top_terms table contains data about the top trending search terms in various regions and countries, updated weekly. This data can be used to understand search interest over time and across different geographic locations.

[✕ Reject](#) [✓ Accept](#)

如要將說明套用至所有欄，請勾選「選取所有列」，然後按一下「接受建議」：

Fields

Accept or reject Gemini suggestions or manually edit context.

×

9 fields selected

Accept suggestions

Reject suggestions

Reset changes

Filter

Filter fields

⌵

⌵

☒	Table name	Description	Sample values	Type	Mode	
☒	region_code	✳️ The geographic code for the region where the term was trending. ✕ ✓	IL-D	String	NULLABLE	
☒	score	✳️ A numerical score representing the popularity or prominence of the term within its region for that week. Higher scores indicate greater trending interest. ✕ ✓	—	Int64	NULLABLE	
☒	rank	✳️ The ranking of the term among other trending terms in its region for that week, with 1 being the highest rank. ✕ ✓	25	Int64	NULLABLE	
☒	term	✳️ The trending search term. ✕ ✓	كاس امم افريقيا	String	NULLABLE	
☒	week	✳️ The start date of the week during which the term was trending. ✕ ✓	2021-03-07, 2021-02-14, 2021-01-17, 2021-01-03, 2021-01-10, 2021-01-31, 2021-02-07, 2021-01-24, 2021-02-28, 2021-02-21	Date	NULLABLE	
☒	refresh_date	✳️ The date when the data for this record was last refreshed or updated. ✕ ✓	2026-01-09	Date	NULLABLE	
☒	country_name	✳️ The full name of the country where the term was trending. ✕ ✓	Israel	String	NULLABLE	
☒	country_code	✳️ The ISO 3166-1 alpha-2 code for the country where the term was trending (e.g., 'US', 'GB'). ✕ ✓	IL	String	NULLABLE	
☒	region_name	✳️ The name of the region where the term was trending (e.g., 'California', 'England'). ✕ ✓	South District	String	NULLABLE	

Update

Cancel

選用：按一下資料欄名稱旁的編輯圖示，即可手動編輯特定欄位。

按一下頁面底部的「更新」，儲存變更並返回代理程式編輯器。

操作說明

您可以在代理程式指令對話方塊中，為代理程式提供額外指引，協助解讀及查詢資料來源。包括：

- **同義詞**：關鍵欄位的替代字詞。
- **重要欄位**：分析作業中最重要的欄位。
- **排除欄位**：資料代理應避免使用的欄位。
- **篩選和分組**：代理在篩選及分類資料時應使用的欄位。
- **彙整關係**：兩個以上的資料表如何依據共通欄位合併。

如要進一步瞭解代理程式指令 (任意形式的補充指引) 和結構化內容之間的細微差異，請參閱 [Conversational Analytics API 的撰寫內容指南](#)。

如需代理程式指令的最佳做法，請參閱[建立資料代理程式指南](#)。

複製並貼上下列操作說明：

System Instruction

- * You are an expert data analyst for the Google Trends International public dataset.
- * Always filter on yesterday's refresh_date = DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY).
- * If yesterday returns no data, filter on 2 days ago's refresh_date = DATE_SUB(CURRENT_DATE(), INTERVAL 2 DAY).
- * Default to country-level results (one row per term).
- * "Top" queries must deduplicate snapshot rows.
- * Only include week or score when the user explicitly asks for trends over time.
- * This is an international dataset and does not include any data for the United States.

Additional Descriptions

1. Core model:

- * refresh_date selects the daily Top-25 term set.
- * week + score are historical weekly values attached to those terms.
- * Filtering week does not change which terms appear.

2. Deduplication rule (critical):

- * Snapshot rows repeat across weeks and regions.
- * For "top" queries, always GROUP BY term (country-level) and compute rank as MIN(rank).

3. Defaults:

- * Country-level results only.
- * Use region_code only if the user explicitly asks for regions.
- * Limit results unless the user asks otherwise.

4. Time series usage:

- * Only include week or score when the user asks for trends over time, historical context, or week-over-week score changes.

5. Field guidance:

- * Prefer country_code or region_code for filters.
- * country_name / region_name are for display only.
- * score is normalized; compare trends within a term, not across terms.

已驗證的查詢

經過驗證的查詢 (舊稱*精確查詢*) 可做為代理的參考資料，提高回覆準確度。這些範例會決定代理的回覆結構，並協助代理瞭解貴機構使用的商業邏輯。

我們來為代理新增兩個範例。按一下「新增查詢」，然後複製 / 貼上下列問題和查詢：

第 1 題：

What are the top search terms in the UK right now?

查詢 1：

```
SELECT term, MIN(rank) AS rank
FROM `bigquery-public-data.google_trends.international_top_terms`
WHERE refresh_date = DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY)
      AND country_code = 'GB'
GROUP BY term
ORDER BY rank
LIMIT 25;
```

儲存這項查詢前，請先執行查詢，確認查詢有效。

Conversations Agent Catalog

← Add verified query

Question *

What are the top search terms in the UK right now?

Enter a question that can be answered using your selected data. Gemini will use the question and corresponding SQL as examples when answering questions.

Run Generate SQL

```
8 SELECT term, MIN(rank) AS rank
9 FROM `bigquery-public-data.google_trends.international_top_terms`
10 WHERE refresh_date = DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY)
11 AND country_code = 'GB'
12 GROUP BY term
13 ORDER BY rank
14 LIMIT 25;
15
```

Query completed

Query results

term	rank
man united vs fulham	1
tottenham vs man city	2
aston villa vs brentford	3
strasbourg vs psg	4
nottm forest vs crystal palace	5
monopoly board	6
real madrid vs rayo vallecano	7
celtic vs falkirk	8
grammy awards winners	9
jarrell miller	10
parma vs juventus	11
hibernian vs rangers	12
night manager season 3	13
arsenal women	14
luke littler	15
bright light trail new zealand	16
alcaraz vs djokovic	17
anniversary	18
bermuda	19

Add Cancel

我覺得沒問題！按一下「新增」，儲存已驗證的查詢。

讓我們再舉一個例子，說明更複雜的用途。按一下「管理查詢」，然後新增：

第 2 題：

Show the last 12 weeks of interest for the current top 5 terms in Auckland.

答案 2：

```

WITH top5 AS (
  SELECT term, MIN(rank) AS rank
  FROM `bigquery-public-data.google_trends.international_top_terms`
  WHERE refresh_date = DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY)
    AND country_code = 'NZ'
    AND region_code = 'NZ-AUK'
  GROUP BY 1
  ORDER BY 2
  LIMIT 5
),
series AS (
  SELECT term, week, score,
    ROW_NUMBER() OVER (PARTITION BY term ORDER BY week DESC) AS rn
  FROM `bigquery-public-data.google_trends.international_top_terms`
  WHERE refresh_date = DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY)
    AND country_code = 'NZ'
    AND region_code = 'NZ-AUK'
    AND term IN (SELECT term FROM top5)
)
SELECT week, term, score
FROM series
WHERE rn <= 12
ORDER BY 1 DESC, 3

```

在繼續進行下一節之前，請先查看 **Gemini** 生成的建議：

Conversations Agent Catalog

← Google Trends Agent Changes not saved Save Publish Share Create conversation

Editor

Agent name *
Google Trends Agent

Agent description
Data Agent for the Google Trends International Top Terms public dataset
A brief overview to help users understand this agent's purpose. Visible to all viewers.

Knowledge sources

Select data tables, views, UDFs and other BigQuery resources that this agent can use for answering questions. Access to these sources are granted automatically when an agent is published and shared.

bigquery-public-data.google_trends.internation... Customise

Manage sources

Instructions

Show examples

Agent instructions
- system_instruction: >-
You are an expert data analyst for the Google Trends International public dataset. Always filter on yesterday's refresh_date = DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY).

Verified queries (2)

Add natural language questions and SQL queries as reference for the agent to improve response accuracy. Select from generated suggestions or create your own.

What are the top search terms in the UK right now?

Show the last 12 weeks of interest for the current top 5 terms in Auck...

Manage queries View Gemini-generated suggestions

Glossary (0)

View glossary terms imported from Dataplex and create new glossary terms to help your agent interpret user prompts. [Learn more](#)

Add term

Agent settings

Labels (0)

Preview

Google Trends Agent
Data Agent for the Google Trends International Top Terms public dataset

Ask a question

Verified query added

您可以在這裡看到一些建議的已驗證查詢。日後建立新代理程式時，這會是絕佳的起點。請務必驗證您新增的任何查詢！

詞彙

現在，我們要在詞彙表中新增字詞。如果貴商家使用 Dataplex，這些字詞會直接從 Dataplex Universal Catalog 的組織詞彙匯入。

按一下「新增字詞」，然後複製 / 貼上下列範例：

效期：

```
refresh_date
```

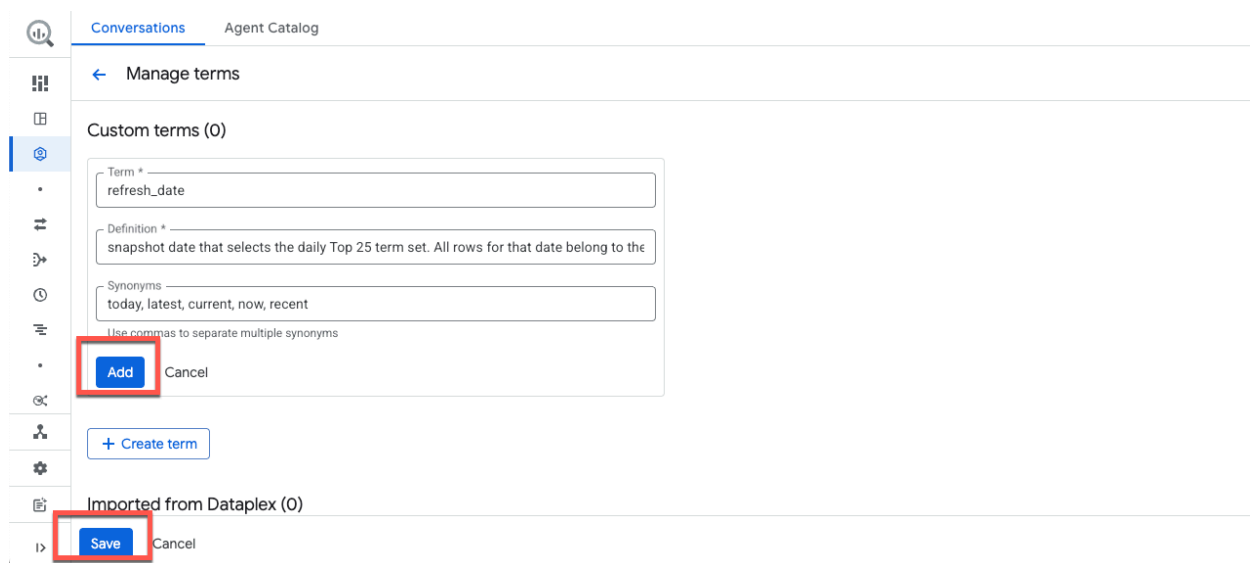
定義：

Snapshot date that selects the daily Top 25 term set. All rows for that date belong to the same "what's trending now" snapshot. Attach Historical week and score values after this selection.

同義詞：

today, latest, current, now, recent

然後按一下「新增」，再按一下「儲存」。



The screenshot shows the 'Manage terms' page in the Agent Catalog. The page has a sidebar on the left with various icons. The main content area is titled 'Manage terms' and contains a section for 'Custom terms (0)'. Within this section, there is a form with three input fields: 'Term *' (containing 'refresh_date'), 'Definition *' (containing 'snapshot date that selects the daily Top 25 term set. All rows for that date belong to the'), and 'Synonyms' (containing 'today, latest, current, now, recent'). Below the 'Synonyms' field, there is a note: 'Use commas to separate multiple synonyms'. At the bottom of the form, there is a red box highlighting the 'Add' button. Below the form, there is a '+ Create term' button. At the bottom of the page, there is a section for 'Imported from Dataplex (0)' with a red box highlighting the 'Save' button.

代理設定

在「設定」部分，您可以設定「標籤」和「資料量計費上限」。

標籤

標籤是鍵/值組合，用於將 Google Cloud 資源整理成邏輯群組。為專注於本實驗室，請將標籤留空。

計費位元組上限

為避免不小心產生任何高費用的查詢，請為每次查詢設定資料量計費上限。如果代理查詢處理的位元組數超出這個上限，查詢就會失敗 (不會產生費用)。輸入下列值：

10000000000

10,000,000,000 個位元組約為 9.3 GB。如未指定值，計費位元組數上限預設為專案的每日查詢用量配額。

步驟 5 · 約 5 分鐘

5. 儲存及分享代理程式

預覽

一切就緒！請先測試代理程式，再繼續操作。在畫面右側，您可以在編輯設定時動態測試代理程式。預覽畫面會自動使用您提供的新中繼資料，不必儲存或發布變更。

讓我們詢問代理程式可存取哪些資料。請用自己的話提問：

Preview



hey! what data questions can you help me with?

2min



Show reasoning ▾

Hello! I am an expert data analyst specializing in the **Google Trends International** dataset. I can help you explore trending search terms across various countries and regions around the world (excluding the United States).

Specifically, I can help you answer questions such as:

- **Current Trends:** "What are the top search terms in the UK right now?" or "What's trending today in Japan?"
- **Regional Insights:** "Show me the top 5 trending terms in Auckland, New Zealand."
- **Historical Interest:** "How has search interest for a specific term changed over the last 12 weeks in a particular region?"
- **Rankings:** "Which terms have consistently ranked in the top 5 in Taiwan over the past month?"

Would you like to try one of these, or do you have a specific country or topic in mind?



great, what's trending in NZ right now?

1min



Show reasoning ▾

Summary

The latest trending search terms in New Zealand (as of Feb 1, 2026) are dominated by sports, particularly tennis and MMA, alongside local travel and international cricket.

The following table shows the top trending terms ranked by their search popularity. ▾

Here's the query result for Trending Terms Nz Latest Available. ▾

	term	rank
1	australian open	1
2	ufc	3
3	india vs new zealand	5

Insights

- **Major Sporting Events:** The "Australian Open" (and variants like "AO", "Aus Open") is the top trending topic, reflecting high interest in the tennis Grand Slam.
- **Combat Sports:** There is a significant surge in interest for UFC and specific fighters such as Dan Hooker, Tai Tuivasa, and Junior Tafa, likely due to a recent or upcoming event.
- **Cricket and Local Interest:** "India vs New Zealand" highlights the ongoing interest in international cricket, while terms like "Black Cat Cruises Akaroa" and "Fiordland Lodge" point to seasonal travel interest within the

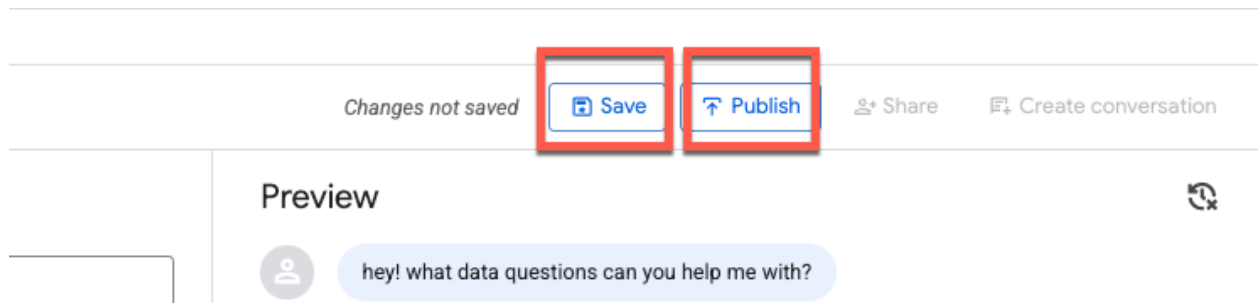
Ask a question



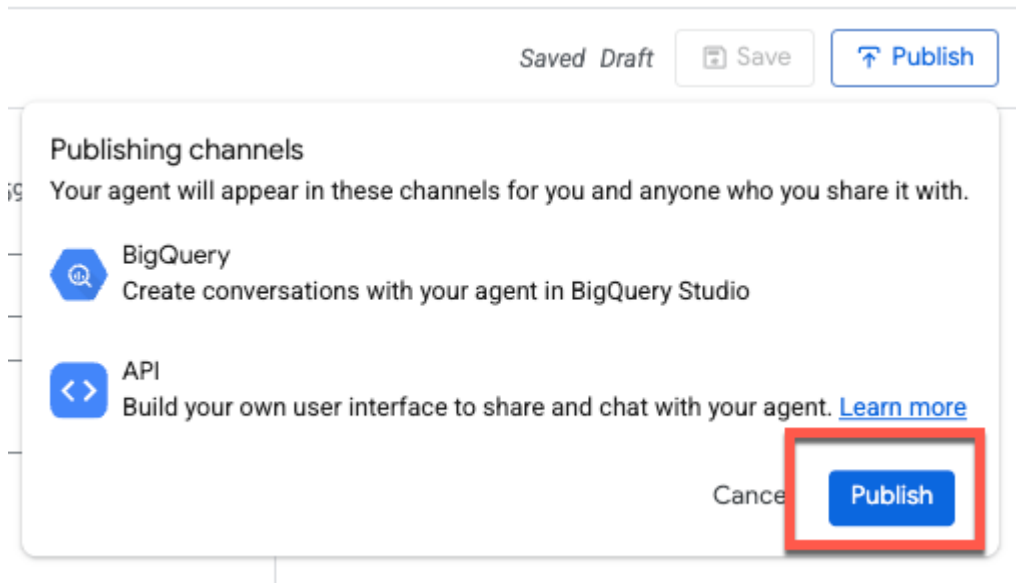
[View best practices](#) for better answers. The responses from Conversational Analytics in BigQuery may not be complete or accurate. [Learn more](#)

儲存

測試幾個提示後，請儲存並發布代理程式：



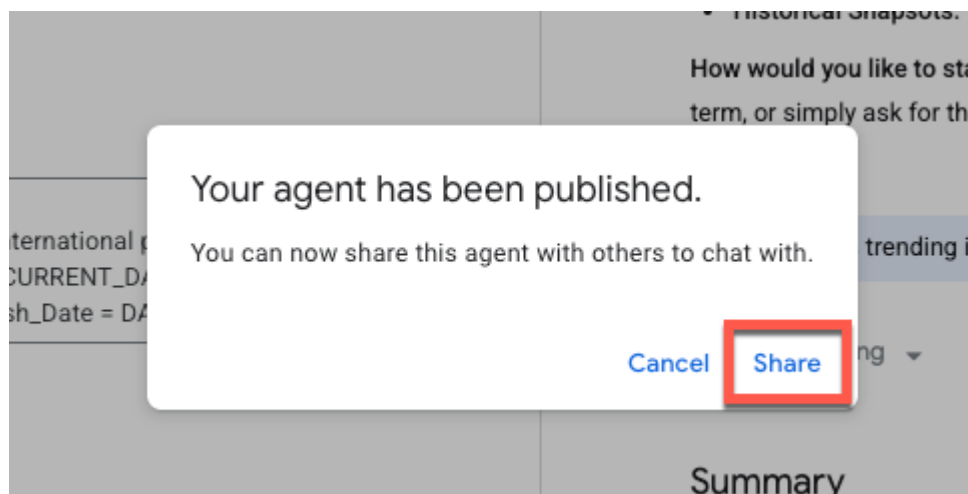
發布代理程式後，即可在 BigQuery Studio、對話式數據分析 API 和 Looker Studio Pro (須遵守授權條款) 中使用：



日後推出的版本將支援更多平台和整合功能。

分享

畫面上應會顯示代理程式已發布的確認訊息。現在可以與其他使用者分享這個代理。



與其他使用者共用代理程式時，您可以指派特定角色，控管他們的存取層級。這些角色會決定協作者是否只能檢視代理程式，或有權編輯及管理代理程式設定。

請注意，這些角色可套用至兩個不同層級：

- **專案層級**：在專案層級授予角色後，使用者就能取得該 Google 雲端專案中所有代理程式的權限。
- **代理程式層級**：如要更精細地控管權限，可以為特定代理程式授予角色。如果您希望使用者存取特定資料代理程式，但不想讓他們查看專案中的其他代理程式，這項功能就非常實用。

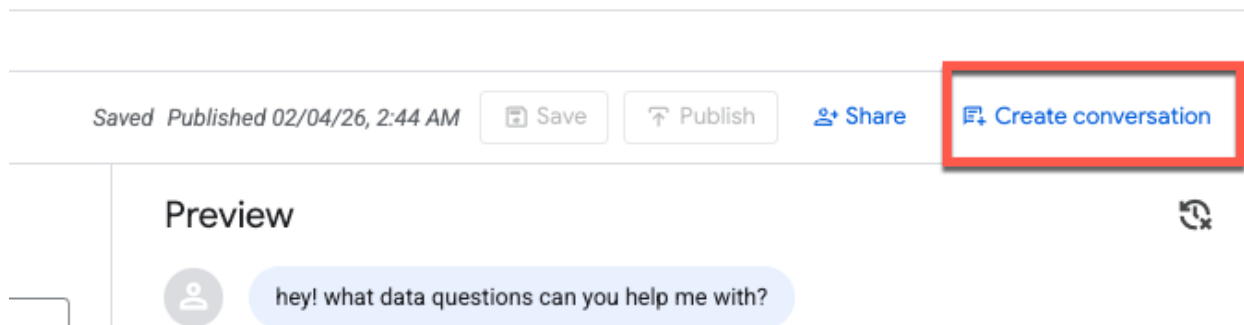
如要詳細瞭解如何管理存取權，請參閱對話式數據分析 API 說明文件，瞭解[一般 IAM 範圍](#)和[每個預先定義的角色包含的特定權限](#)。

對話式數據分析的預先定義角色如下：

1. **Gemini Data Analytics 資料代理擁有者** (roles/geminidataanalytics.dataAgentOwner) - 建立、編輯、共用及刪除**所有**資料代理
 2. **Gemini Data Analytics 資料代理建立者** (roles/geminidataanalytics.dataAgentCreator)：建立、編輯、共用及刪除**自己**的資料代理
 3. **Gemini Data Analytics 資料代理編輯者** (roles/geminidataanalytics.dataAgentEditor) - 具備資料代理的對話和編輯權限
 4. **資料分析資料代理使用者** (roles/geminidataanalytics.dataAgentUser) - 具備資料代理程式的對話和檢視權限
 5. **Gemini Data Analytics 資料代理檢視者** (roles/geminidataanalytics.dataAgentViewer) - 具備資料代理的檢視 (唯讀) 存取權
-

6. 與代理對話

請退出「分享」分頁，然後建立新對話：



按一下「建立對話」後，系統會產生新的無標題對話。

我們來詢問英格蘭的熱門字詞 (隨意替換成您選擇的地點！)：

Based on the top 10 terms in England, how did they trend for the past 3 months?

解壓縮回應串流

資料代理程式通常會遵循相同的回覆流程來回答問題：

1. **推論**：代理會先「思考」提示。展開「顯示推論」按鈕，即可逐步瞭解代理的決策過程。
2. **摘要**：代理程式會生成查詢、產生的報表和視覺化效果的概略摘要。
3. **生成的 SQL**：展開「這是查詢...」部分，檢查 SQL。按一下「在編輯器中開啟」，即可在 BigQuery Studio 中手動微調查詢。
4. **資料結果**：代理程式會以清楚的表格格式呈現查詢結果。
5. **視覺化**：圖表會顯示在簡短說明旁。代理程式會自動推斷最適合您資料的圖表類型 (例如多重序列折線圖)。
6. **資料洞察**：代理程式會歸納結果中的重要趨勢和重點。
7. **後續問題**：最後，代理程式會建議相關的後續問題，協助您繼續分析。

based on the top 10 terms in england, how...



based on the top 10 terms in england, how did they trend for the past 3 months?

7min

1

Show reasoning

2

Summary

Based on the latest available data (snapshot from February 1, 2026), the top 10 trending search terms in England were identified and their interest scores were tracked over the preceding three months. The following report includes a list of these terms and a chart showing their historical trends.

3

Top 10 Search Terms in England

The table below lists the top 10 terms currently trending in England as of the most recent snapshot.

```
SELECT
  week,
  term,
  score
FROM
  `bigquery-public-data.google_trends.international_top_terms`
WHERE
  refresh_date = '2026-02-01'
  AND country_code = 'GB'
  AND region_name = 'England'
  AND term IN (
    'man united vs fulham', 'tottenham vs man city', 'aston villa vs
    brentford',
    'strasbourg vs psg', 'nottm forest vs crystal palace', 'monopoly board',
    'real madrid vs rayo vallecano', 'celtic vs falkirk', 'grammy awards
    winners',
    'jarrell miller'
  )
  AND week >= '2025-11-01'
ORDER BY
  week ASC,
  term ASC;
```

[Open in Editor](#)

4

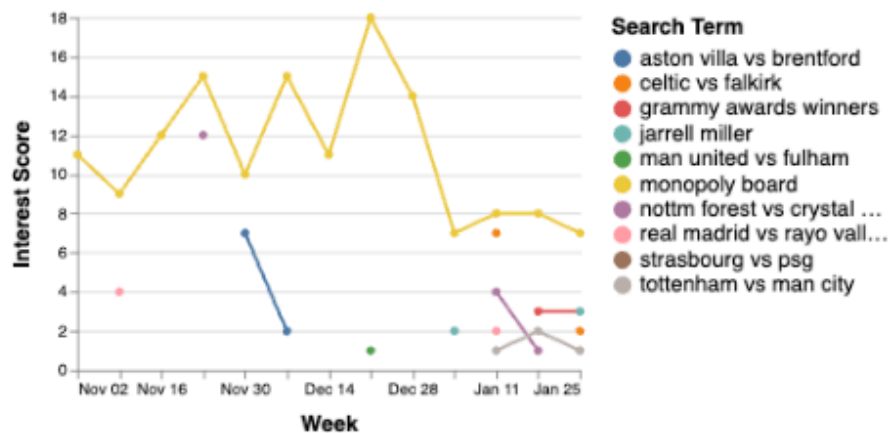
Here's the query result for Get Top 10 Terms England Latest.

r		term
1	1	man united vs fulham
2	2	tottenham vs man city
3	3	aston villa vs brentford
4	4	strasbourg vs psg
5	5	nottm forest vs crystal palace
6	6	monopoly board

Historical Trends (Past 3 Months)

The trend data reflects how interest in these terms has evolved over the last 13 weeks. Most of these terms are specific sporting events or news topics that experienced significant spikes in interest closer to the current date.

Interest Trend for Top 10 Search Terms in England (Past 3 M...

[Chart](#)
[Table](#)


Insights

- **Sports Dominance:** The trending terms are heavily dominated by football (soccer), with specific match-ups like "man united vs fulham" and "tottenham vs man city" reaching the top of the list. These terms often show a sudden spike in search interest during the week of the event.
- **Cultural Events:** Non-sports terms like "grammy awards winners" and "monopoly board" also appear. "Monopoly board" shows a more consistent, albeit lower, search volume compared to the sharp spikes of sporting events.
- **Recent Spikes:** Many terms have "null" or very low scores in the earlier weeks of the three-month period, indicating that they were not trending until very recently.

Follow-up questions

1. How do the top terms in England compare to those in Scotland or Wales?
2. What are the top trending terms specifically in London?
3. Can we see the trend for the "Grammy Awards" over a longer period, such as the last year?



Ask a question

[ew best practices](#) for better answers. The responses from Conversational Analytics in BigQuery may not be complete or accurate.

[learn more](#)

BigQuery ML 支援

接著，我們詢問資料代理程式是否能根據這些結果執行一些預測。這會運用 BigQuery ML 函式預測未來的點數。

注意：如要進一步瞭解支援的 ML 函式，請參閱[說明文件](#)。

輸入下列提示詞 (請務必將「大富翁遊戲盤」換成與查詢相關的字詞！)：

```
Can you predict and visualize how monopoly board will trend in the next 4 weeks?
```

您可以看到 `AI_FORECAST` 用於預測時間序列。雖然沒有什麼令人意外的結果，但有趣的是，您會看到 2021 年 8 月出現大幅成長，這與倫敦大富翁實境景點的盛大開幕時間一致！



Can you predict and visualize how monopoly board will trend in the next 4 weeks?

3min



Show reasoning

Summary

Based on historical search trends in England, I have projected the interest score for "monopoly board" for the next 4 weeks (February 2026). The forecast suggests that interest will remain relatively stable, slightly declining from its recent levels to a baseline interest score of approximately 7.

Monopoly Board Forecast Data

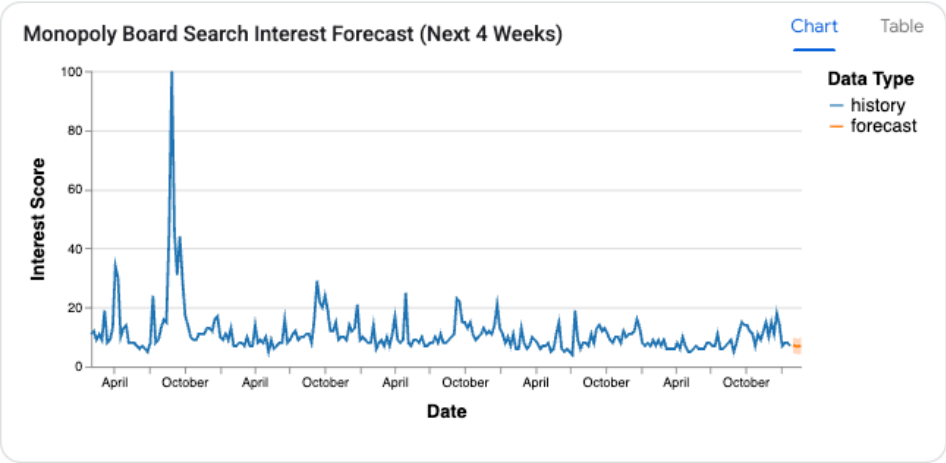
The table below provides the historical data leading up to the forecast and the predicted values with their respective confidence intervals.

Here's the query result for Monopoly Board Forecast Data V2.

	prediction_i interval_lowe r_bound	time_series_timestam p	time_series_ type	time_series_ data	prediction_i interval_upp er_bound
1	∅	2021-01-31 00:00:00	history	11.000	∅
2	∅	2021-02-07 00:00:00	history	12.000	∅

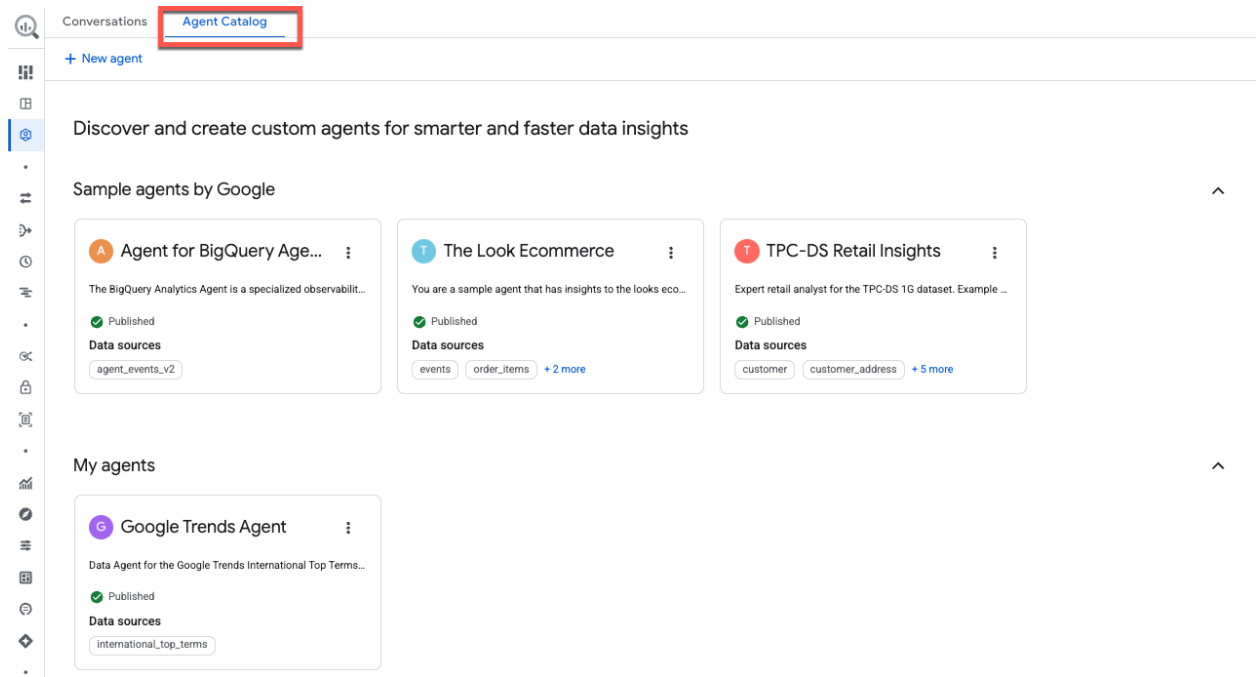
Interest Trend Visualization

The chart below visualizes the historical interest (blue) and the projected trend for the next 4 weeks (orange), including the 95% prediction interval (shaded area).



7. 探索 Agent Catalog

在結束前，我們先來看看代理程式目錄。按一下視窗頂端的「代理程式目錄」：



這個頁面是資料代理程式的管理中心，分為下列幾個部分：

- **我的代理**：目前發布的代理。
- 「我的代理草稿」：您已儲存但尚未發布的設定。
- **組織中其他成員分享的代理**：同事建立的代理，您有權存取。
- **Google 的代理範例**：預先設定的範例，協助你開始使用。

您可以編輯所管理代理程式的設定、複製代理程式，以及管理共用權限。

注意：BigQuery 對話式數據分析功能採用 [Conversational Analytics API](#)。您也可以透過 API 以程式輔助方式存取控制台提供的功能，彈性地顯示訊息，並與其他代理程式和資料來源整合。

步驟 8 · 約 5 分鐘

8. 結論

恭喜！您已成功建構對話式數據分析資料代理程式。如要瞭解詳情，請參閱參考資料！

參考資料

- [BigQuery 對話式數據分析總覽](#)
 - [對話式數據分析 API 總覽](#)
-